**Organic Cumulative Examination**

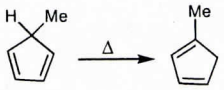
February 9, 2018

This cume has 20 points and 14 points or more are needed to pass

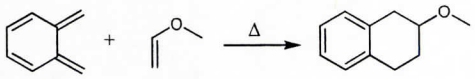
Examine the following pericyclic reactions. For each reaction, indicate, whether it is an electrocylic reaction, a cycloaddition reaction or a sigmatropic rearrangement. (3 points)

a)

b)




c)




2) The molecule to be examined is (2E, 4Z, 6Z, 8E)-decatetraene (8 points)

a) Under thermal conditions, will ring closure be conrotatory or disrotatory?

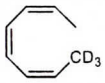
b) Will the product have cis or trans configuration?

c) Under photochemical conditions, will ring closure be conrotatory or disrotatory?

d) will the product have the cis or the transconfiguration?

3) Give an example for a Claisen rearrangement reaction (4 points)

4) Account for the differences in the products obtained under photochemical and thermal conditions for the transformation of




Give the reaction schemes including arrows for electron movement under both reaction conditions (5 points)